

36 Organic synthesis

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Plan retrosynthetically from the target, identify the key bond/functional group, then reverse to feasible starting materials. Forward route needs reagents, solvents and conditions; include isolation and purification. Percentage yields compound across steps and can make short routes preferable even if individual reactions look efficient.

Worked example: follow the reasoning

An overall two-step route with 90% then 80% yields gives 72% overall, compared with $95\%^3 \approx 85.7\%$ for a three-step route; length alone does not determine success.

Quick check · multiple choice

A What is a necessary check in retrosynthetic route planning? [1]

- A Ignore carbon count
- B Only estimate colour
- C Never select conditions
- D Account for functional-group changes

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A three-step synthesis gives 80% yield at each step. Calculate overall yield and compare with a one-step 50% yield. [3]

Final answer: _____

Answers and explanation

Quick check: D. Account for functional-group changes

Question 1 [3 marks]

$0.8^3 = 0.512$ or 51.2% [2]; three-step route has slightly greater overall yield than 50% [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.